21 Lecture 21 (March 5, 2025, est.): Nondegeneracy and Frobenius injectivity

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Let k be algebraically closed of characteristic $p > 0$ and let $\mathcal{C}$ be a symmetric fusion category. We continue to use *nondegenerate* for $\dim(\mathcal{C}) \neq 0$.

21.1 Nondegenerate categories have super fiber functors

21.1 Proposition. *If $\mathcal{C}$ is nondegenerate, then it admits a symmetric tensor functor to* Vect *when $p = 2$, and to* SVec *when $p > 2$.*

Proof. Use Theorem 20.5 to lift $\mathcal{C}$ symmetrically over $W(k)$. Extend the generic fiber to an algebraic closure of $K = \operatorname{Frac} W(k)$. By Deligne's theorem for symmetric fusion categories in characteristic zero, the result is $\operatorname{Rep}_{\overline{K}}(G, z)$ for a finite group G and a central element z with $z^2 = 1$. Here z acts as parity and the symmetry is the super symmetry. Its global dimension is $|G|$: the superdimensions of simples differ from their ordinary dimensions only by signs, which disappear on squaring.

Global dimension of the integral lifting reduces to $\dim(\mathcal{C})$. Thus $|G|$ is not divisible by p. The category $\operatorname{Rep}_k(G, z)$ is consequently semisimple and nondegenerate, and its standard integral lifting has the same geometric generic fiber. Faithfulness of lifting, [ENO05, Corollary 9.9], identifies $\mathcal{C}$ with $\operatorname{Rep}_k(G, z)$. The use of the geometric generic fiber avoids assuming that the fraction field K is algebraically closed. For $p > 2$, take the forgetful functor to SVec. If $p = 2$, the group has odd order, so $z = 1$ and the forgetful functor lands in Vect. ∎

21.2 A dimension test for full faithfulness

For a tensor functor $F : \mathcal{C} \to \mathcal{D}$ between fusion categories, write $F(\mathcal{C})$ for its *fusion image*: the full subcategory closed under direct sums and direct summands generated by the objects $F(X)$. This need not be merely the list of objects literally equal to $F(X)$.

21.2 Lemma. *One has*

$$\operatorname{FPdim}(F(\mathcal{C})) \leq \operatorname{FPdim}(\mathcal{C}),$$

with equality if and only if F is fully faithful.

Proof. Replace the target by the fusion image $\mathcal{D}_0 = F(\mathcal{C})$; then F is dominant. Let R be its right adjoint. The standard dimension identity for a dominant tensor functor is

$$\operatorname{FPdim}(\mathcal{C}) = \operatorname{FPdim}(\mathcal{D}_0) \operatorname{FPdim}(R(\mathbf{1})).$$

For completeness, if N_{ia} is the multiplicity of the simple Y_a in $F(X_i)$, the vector $\sum_i \operatorname{FPdim}(X_i) N_{ia}$ is the image of the regular dimension vector of $\mathcal{C}$. It is therefore an eigenvector for multiplication by $F(\bigoplus_i X_i)$. Dominance means that this object contains every simple of $\mathcal{D}_0$; its multiplication matrix has strictly positive entries,

by rigidity. Perron–Frobenius uniqueness makes the displayed vector a multiple of $\mathrm{FPdim}(Y_a)$; its coefficient at the unit is $\mathrm{FPdim}(R(\mathbf{1}))$. Summing against $\mathrm{FPdim}(Y_a)$ gives the identity. The unit occurs once in $R(\mathbf{1})$ by adjunction, so this last dimension is at least one. Equality holds exactly when $R(\mathbf{1})$ is the unit. The projection formula $R(F(X)\otimes Y)\simeq X\otimes R(Y)$, obtained by rigidity and adjunction, then gives $RF\simeq \mathrm{id}$, which is equivalent to full faithfulness. Conversely a fully faithful F gives $R(\mathbf{1})\simeq\mathbf{1}$. ∎

21.3 Definition. The category $\mathcal{C}$ is *Frobenius injective* if $\mathrm{Fr}:\mathcal{C}\to\mathcal{C}^{(1)}\boxtimes \mathsf{Ver}_p$ is fully faithful.

The word "injective" in this definition means fully faithful, not merely faithful. Every tensor functor under consideration is already faithful.

21.3 A Grothendieck-ring criterion

21.4 Lemma. *If the finite-dimensional algebra $A=K_0(\mathcal{C})\otimes_{\mathbb{Z}} k$ is semisimple, then $\mathcal{C}$ is nondegenerate.*

Proof. Let $t(a)$ be the trace of left multiplication by a on A. Since A is commutative semisimple over the algebraically closed field k, it is a product of copies of k, and $(a,b)\mapsto t(ab)$ is nondegenerate. Set $r=\sum_Z[Z^*\otimes Z]$, summing over the simples. For simple X,Y, rigidity gives

$$t([X][Y])=\sum_Z \dim_k \mathrm{Hom}(X\otimes Y\otimes Z,Z)=\dim_k\mathrm{Hom}(X\otimes Y,\bigoplus_Z Z^*\otimes Z),$$

with the integer on the right read in k. The matrix of multiplication by r therefore agrees with the matrix of this trace pairing after permuting the basis by duality. It is invertible, so r is a unit of A. Apply the categorical-dimension homomorphism $A\to k$: the image of r is $\sum_Z \dim(Z)^2=\dim(\mathcal{C})$, which must be nonzero. ∎

21.5 Lemma (Type Vect)**.** *A Frobenius-injective category of Frobenius type* Vect *is nondegenerate.*

Proof. Here $\mathrm{Fr}:\mathcal{C}\to\mathcal{C}^{(1)}$ is fully faithful. The two categories have the same Frobenius–Perron dimension, so this is an equivalence. Equation (20.1) shows that taking pth powers permutes the simple-object basis of A, with the Frobenius automorphism on scalar coefficients. Thus the map $a\mapsto a^p$ is bijective. It is Frobenius-semilinear, rather than k-linear, but bijectivity still follows because k is perfect. An element killed by some power p^m must therefore be zero. Hence A is reduced, and a finite-dimensional commutative reduced algebra over k is semisimple. Apply Lemma 21.4. ∎

21.6 Lemma (Type SVec)**.** *Suppose $p>2$. A Frobenius-injective category of Frobenius type* SVec *is nondegenerate.*

Proof. Full faithfulness makes the image of a simple object simple. Thus for every simple X, either $\mathrm{Fr}(X)=Y\boxtimes\mathbf{1}$ or $\mathrm{Fr}(X)=Y\boxtimes L_{p-1}$. These alternatives give a faithful $\mathbb{Z}/2$-grading $\mathcal{C}=\mathcal{C}_0\oplus\mathcal{C}_1$. The neutral component has Frobenius type

Vect, and its Frobenius functor is fully faithful by naturality. Lemma 21.5 gives $\dim(\mathcal{C}_0) \neq 0$.

Here is the dimension calculation for the grading. Put $r_i = \sum_{X \in \mathcal{C}_i} \dim(X)[X]$ in $K_0(\mathcal{C}) \otimes k$. For a simple $Y \in \mathcal{C}_1$, the fusion rules and rigidity imply $[Y]r_0 = \dim(Y)r_1$; this follows by comparing the coefficient of each simple, using $N_{Y,X}^{Z} = N_{Y^*,Z}^{X}$. Apply categorical dimension and cancel $\dim(Y) \neq 0$ to obtain $\dim(\mathcal{C}_0) = \dim(\mathcal{C}_1)$. Consequently

$$\dim(\mathcal{C}) = 2\dim(\mathcal{C}_0) \neq 0.$$

■

21.4 The other Frobenius types

21.7 Lemma. *Suppose $p > 3$ and $\mathcal{C}$ has Frobenius type Ver_p^+ or Ver_p. Then its Frobenius fusion image contains $\mathbf{1} \boxtimes \mathsf{Ver}_p^+$.*

Proof. Some $\mathrm{Fr}(X)$ contains a simple summand $Y \boxtimes L_r$ with $r \neq 1, p-1$. Its tensor product with its dual contains $(Y \otimes Y^*) \boxtimes (L_r \otimes L_r^*)$. The first factor contains the unit, and the fusion rules show that the second contains L_3. Thus the fusion image contains $\mathbf{1} \boxtimes L_3$, which generates $\mathbf{1} \boxtimes \mathsf{Ver}_p^+$. ■

This lemma does not require Frobenius injectivity. Full faithfulness will be used in the next lecture to lift individual summands back to $\mathcal{C}$ and complete the proof of Ostrik's theorem.

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